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Project Title:	Therapeutic Modulation of Systemic Glucose-Dependent Insulintropic Peptide Levels to Counteract Microgravity-Induced Bone Loss		
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Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
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Contact Monitor:		**Contact Phone:**	
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	(1) Original Aims: Our proposal contained two specific aims: (1) Determine whether elevations in endogenous Glucose-dependent insulinotropic peptide (GIP) levels lead to an increase in bone formation by examining the bone phenotype in GIP-overexpressing transgenic mice and (2) Determine whether endogenous elevations in GIP prevent bone loss in a model of weightlessness by measuring bone turnover rate in GIP-overexpressing transgenic mice with hindlimb suspension. (2) Key findings of the project In an effort to define the role of GIP in normal bone formation three types of experiments were performed: (a) studies		

Task Description:	characterizing the GIP-overexpressing Tg mice (b) studies characterizing the GIPR KO mouse and (c) studies characterizing the potential interactions between GIP and another anabolic agent such as PTH. (a) Studies characterizing the GIP-overexpressing transgenic mice As an initial strategy we developed transgenic mice overexpressing GIP to examine the bone phenotype in the transgenic lines. The transgenic animals had a significantly higher bone density. We next evaluated the protective effects of GIP on the bone loss associated with estrogen loss. 24 three-month old mice were divided into four equal groups and received 25mM ZnSO4 supplemented in their drinking water for 8 weeks. We found that the GIP overexpressing transgenic mice were protected against the bone loss associated with ovariectomy. (b) Studies characterizing the GIP receptor knockout mouse A GIP receptor knockout mouse (GIPRR KO) model was developed by Miyawaki et. al. in order to determine whether abnormalities in GIP secretion played a role in the development of Type II diabetes mellitus. These investigators had no interest in examining the bones of these animals and we have obtained these GIP receptor knockout mice from Japan. GIPR KO mice have a 39% lower BMC than control mice at one month of age, however at three months of age GIPR KO BMC was only 11% lower than control mice and only 12% lower at five months of age. The markers for bone formation (alkaline phosphatase and osteocalcin) were significantly lower in GIPR KO than control mice. In contrast, the marker of bone breakdown (pyridinium cross-links) was not significantly different in GIPR KO vs control mice. These data would suggest that the main reason that the bone mass of GIPR KO mice is lower than control mice is because of decreased bone formation, rather than increased bone breakdown. Taken together these data on the GIP Tg and GIPR KO mice strongly support our hypothesis that GIP plays an important role in normal bone formation. (c) Studies characterizing the potential interactions between GIP and another anabolic agent such as PTH Markers of bone breakdown have been shown to drop on feeding, consistent with inhibition of osteoclastic activity. GIP effects on bone resorption were examined and GIP was found to be a powerful inhibitor of bone resorption. Thus, we hypothetized that GIP, when combined with PTH might have an additive or synergistic effect on bone formation. The GIP and PTH doses selected were low so as to be able to detect potential synergistic interactions with the combined treatment. Osteocalcin measurements were made just at the end of the one month experiment and all treatment groups were found to be significantly different from control. We also examined alkaline phosphatase at the intermediate point (after two-weeks of treatment and found that only the combination of GIP + PTH had a significant effect on raising alkaline phosphatase. To further assess the interactions between GIP and PTH we examined the effect of these agents on bone strength. Interestingly, the combination of GIP and PTH resulted in a significant increase in bone stiffness with no effect on Young's modulus. Further experiments are proposed to further define the differences between GIP and PTH action. Nevertheless, these experiments demonstrate the potential for the combination of GIP + PTH to have additional beneficial effects on bone mass and on the prevention of microgravity-induced bone loss. (3) The impact of these findings on the hypotheses objectives and specific aims of the original proposal The findings of the past year validate our hypothesis that GIP could be a useful countermeasure to microgravity-induced bone loss. Our data is consistent with GIP being the hormone linking nutrient intake and bone formation. (4) The proposed research plan for the coming year We have obtained an oral inhibitor of the enzyme that breaksdown GIP, dipeptidyl dipeptidase IV (DPP-IV). We propose to administer this agent to mice in order to determine whether the GIP elevations induced by this agent are sufficient to mimic our findings of the increased bone mass in GIP-overexpressing transgenic mice.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a disease characterized by an imbalance between bone formation and breakdown leading to a decrease in bone mass, resulting in an increased risk of fractures. Osteoporosis is associated with significant morbidity and mortality due to fractures, in particular vertebral compression fractures and hip fractures. A number of epidemiological and clinical studies have linked abnormalities in nutrition to osteoporosis. Thus, although there is a considerable amount of clinical data to link abnormalities in nutritional intake with osteoporosis, very little research has delved into this phenomenon. Present therapy for osteoporosis is predominantly antiresorptive therapy, i.e. to prevent further bone breakdown and few medications (except for parathyroid hormone injection, perhaps estrogen in women and fluoride) are able to increase new bone formation. Our studies should shed further light on the mechanisms involved in the development of osteoporosis, and thus help in the development of new anabolic therapies.
Task Progress:	Major new findings during this past year of funding include: (1) Validation of our in vitro findings suggesting a role for GIP as a link between nutrient intake and bone formation. We now have two animal models: (a) a GIP-overexpressing transgenic mouse which has a higher bone mass which, the indices demonstrate, have higher bone formation and decreased bone breakdown (b) both a functional GIP receptor knockout mouse, generated by chronically exposing them to high GIP levels, and an actual GIP receptor knockout mouse. Both of these mice have a lower bone mass, higher indices of bone breakdown and lower indices of bone formation. These mice are more prone to bone loss. These data suggest that GIP levels may be an important risk factor for development of osteoporosis and validates our original proposal that GIP might be a valuable countermeasure to microgravity-induced bone loss. (2) Demonstration of GIP as a major nutritional "integrative" hormone capable to affect multiple tissues to maximize nutrient utilization. Thus, manipulation of GIP for bone maintenance in astronauts in space may have the additional benefit of increasing nutrient utilization and maximize nutritional health.

Bibliography Type:	Description: (Last Updated:)
Articles in Peer-reviewed Journals	Zhong Q, Sridhar S, Ruan L, Ding KH, Xie D, Insogna K, Kang B, Xu J, Bollag RJ, Isales CM. "Multiple melanocortin receptors are expressed in bone cells." Bone, in press, June 2004. , Jun-2004
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